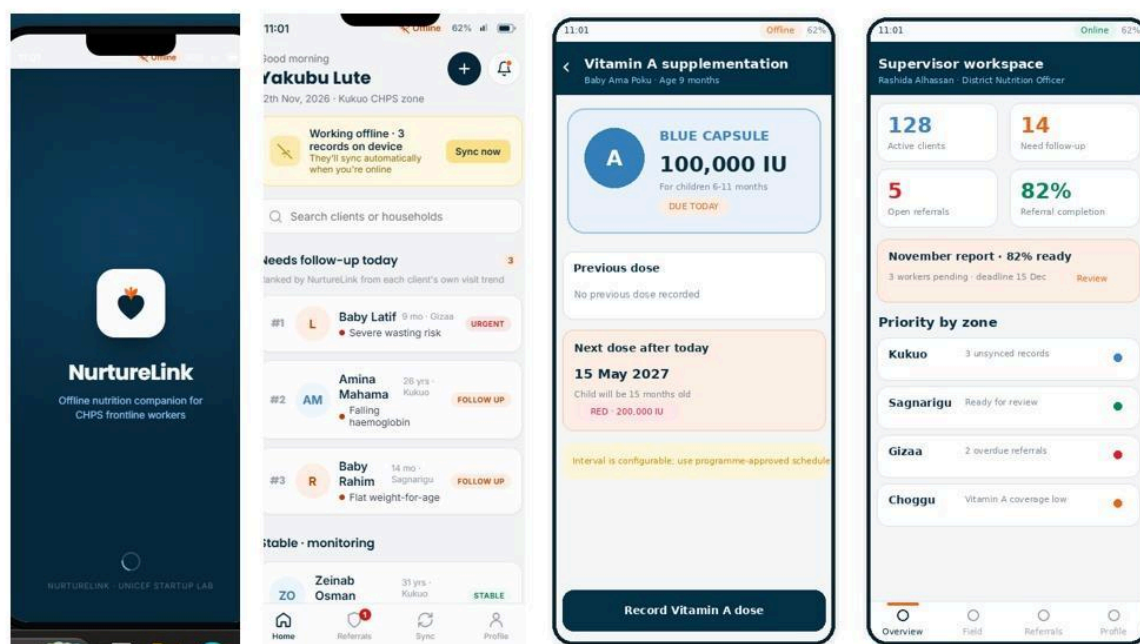


NurtureLink

Step-by-Step Prototype Review and Developer Recommendations

Health-worker and supervisor interfaces

Problem statement: Develop a practical, low-connectivity, community-level AI solution that empowers frontline health workers and caregivers to accelerate maternal and under-five survival in Northern Ghana.



Prepared as a clear implementation guide for the development team

How to Use This Document

This document combines all recommendations made while reviewing the health-worker and supervisor prototypes, together with practical Child Welfare Clinic experience. **Each major recommendation is followed by a suggested app interface so you can see how the idea should look in practice.**

The main idea is simple: NurtureLink should show what is due, what is changing, what is dangerous, and what action must happen next.

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1. What Is Already Strong

I think the prototype is already strong after reviewing both the health-worker and supervisor dashboards. The offline feature, local-food feeding plans, voice counselling, shared-device login, explainable client ranking, referral follow-up and district overview are very practical for CHPS work.

The app is already doing more than storing information. It helps a frontline worker answer four useful questions:

- Who needs attention first today?
- Why is this mother, newborn or child at risk?
- Should I counsel, monitor or refer the client?
- Did the counselling, referral or follow-up actually happen?

The goal is not to replace the health worker. The app should organize information, explain risk and support the next action. The trained health worker remains responsible for the final decision.

2. Product Goal and Design Rules

NurtureLink should work like a digital family health folder that continues working when there is no internet. It should follow a family across pregnancy, delivery, breastfeeding, newborn care, immunization, nutrition and child growth.

Pregnancy → Delivery → Postpartum/lactating mother → Newborn → Infant feeding → Child growth monitoring

Every feature should support at least one of the following jobs: early danger detection, household prioritization, caregiver guidance, referral completion, routine CWC work, data quality, or monthly reporting.

- **Offline first:** The essential workflow must work without internet.
- **Explainable:** The app must show why a client was ranked or referred.
- **Human reviewed:** The health worker approves counselling and referral actions.
- **Client specific:** Pregnant women, lactating mothers, newborns and children must not use one mixed form.
- **Safe:** Danger signs should bypass ordinary counselling and trigger urgent action.
- **Configurable:** Clinical thresholds, vaccine schedules and DHIMS2 deadlines must be updateable without rebuilding the app.

3. Navigation and Role Clarity

3.1 Show the active tab clearly

At the moment, it is difficult to tell quickly whether the user is on Home, Referrals, Sync, Profile, Overview or Field. A selected tab should look selected.

- Use a filled icon for the active tab.
- Make the active label bold.
- Use a stronger colour.
- Add an orange line, dot or soft background pill.
- Keep the selected tab visible when the user opens a related detail page.

Example: Home uses a filled navy icon, bold text and an orange line. The other tabs use outline icons and lighter text.

3.2 Keep the workspace role visible

- Health-worker workspace — Yakubu Lute · Kukuo CHPS Zone
- Supervisor workspace — Rashida Alhassan · District Nutrition Office

When Rashida opens Yakubu's caseload, the screen should not look as if Rashida has become Yakubu. It should clearly show:

Viewing Yakubu Lute's caseload — Read-only supervisor view

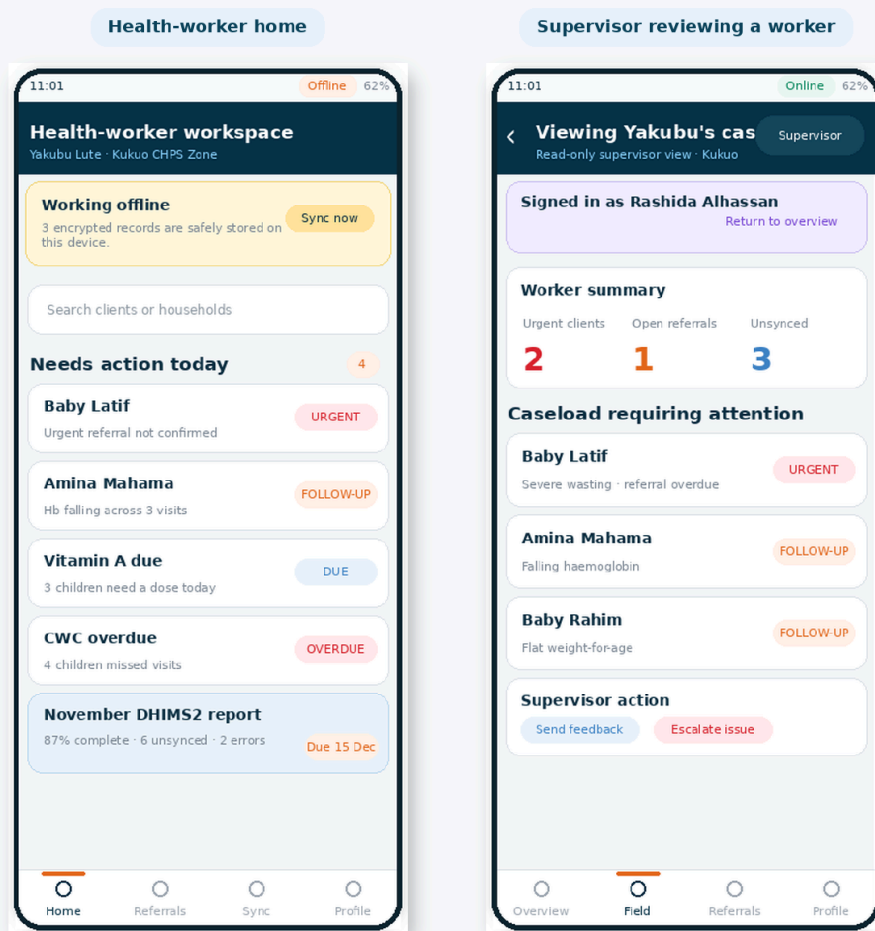
- Add a clear "Return to district overview" button.
- Keep the supervisor banner visible while reviewing a worker's records.
- Do not allow silent editing of a worker's clinical record.

Suggested Interface: Navigation and Role Clarity

The active tab and the user role should remain obvious at every point in the app.

IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.



DEVELOPER FOCUS

- Use a filled icon, bold label and orange marker for the selected tab.
- Keep "Health-worker workspace" or "Supervisor workspace" visible.
- A supervisor reviewing a worker must see a read-only banner and a clear return button.

4. Household, Client and Folder Identification

Names alone are not enough. Two people can have the same name, and one person may be registered twice with slightly different spelling. NurtureLink should identify the household and each person inside it.

Identifier	How it should work
Household ID	One number that links people living in the same household. Example: HH-KUK-0147.
NurtureLink Client ID	A permanent ID generated automatically, even offline. Example: NL-KUK-2026-00428.
Existing health-system ID	ANC folder number, CWC card number, facility folder number or Maternal and Child Health Record Book number, when available.
Location details	Community, landmark, phone number when available, and Ghana digital address where available.

4.1 Link mother and child records

Amina Mahama — mother | Baby Rahim — child | Household HH-KUK-0147

After delivery, the app should link the newborn to the mother’s pregnancy record. This prevents the mother’s pregnancy history from disappearing when the baby is born.

4.2 Check for duplicate clients before saving

The offline duplicate check should compare:

- Name and close spelling matches
- Date of birth or estimated age
- Community
- Mother or caregiver name
- ANC/CWC/facility folder number
- Household ID

Possible existing record found: Ama Poku · Kukuo · Household 0182. Review the existing record before creating a new client.

The same control is needed for referrals. Baby Latif appeared more than once with what looked like the same active referral. The app should ask the worker whether to update the existing referral or create a genuinely new referral episode.

5. Client Pathways and Registration Forms

The app should support four connected client stages:

1. Pregnant woman

2. Postpartum or lactating mother
3. Newborn
4. Child under five

5.1 Pregnant-woman registration

- Full name
- Date of birth or age
- Community
- Household ID
- Phone number or household landmark
- Ghana digital address, when available
- ANC folder number
- Last menstrual period, when known
- Expected delivery date
- Gestational age, calculated automatically
- Gravida and parity
- Main ANC facility
- Client consent

For a pregnant adult, use “Client consent given,” not “Caregiver consent given.”

5.2 Child registration

- Child’s full name
- Sex
- Date of birth or estimated age
- Mother or caregiver name
- Relationship to the child
- Community
- Household ID
- Phone number or household landmark
- Ghana digital address, when available
- CWC card number
- Caregiver consent

5.3 Newborn creation after delivery

Amina has delivered. Create and link newborn record?

The app should allow the worker to create the newborn from the mother’s record so that the mother and baby are connected from the beginning.

Suggested Interface: Household Search and Registration

Search first, avoid duplicates, then collect different information for a child and a pregnant woman.

IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Search before registration

11:01Offline62%

< Find household or client

Search before registration

SEARCH

Name, client ID, folder/card number, household ID, phone, landmark or digital address

Search local records

POSSIBLE MATCH

Ama Foku - Child - Kukuo
CWC-KUK-0214 - HH-KUK-0182
DOB: 4 Jan 2025 - Caregiver: Akosua

Open recordNot same person

NO MATCH?

Create a new household or client only after checking the possible matches above.

Create new record

HomeReferralsSyncProfile

Child registration

11:01Offline62%

< Register child

Child pathway

CHILD DETAILS

FULL NAME

Child's name

SEX

Select sex

DATE OF BIRTH

DD/MM/YYYY

ESTIMATED AGE

If DOB unknown

CWC CARD NUMBER

Enter if available

CAREGIVER AND HOUSEHOLD

CAREGIVER NAME

Mother or caregiver

RELATIONSHIP

Select

PHONE

Optional

HOUSEHOLD

HH-KUK-0147Linked

LOCATION

Community, landmark or digital address

☐ Caregiver consent given

NurtureLink ID will be generated offline

Save child and start CWC visit

Pregnancy registration

11:01Offline62%

< Register pregnant woman

Maternal pathway

IDENTITY

FULL NAME

Woman's name

AGE / DOB

Enter

PHONE

Optional

HOUSEHOLD

HH-KUK-0147Linked

PREGNANCY DETAILS

ANC FOLDER NUMBER

Enter if available

LAST MENSTRUAL PERIOD

If known

EXPECTED DELIVERY

DD/MM/YYYY

GESTATIONAL AGE

24 weeksAuto

GRAVIDA

G

PARITY

P

MAIN ANC FACILITY

Select facility

☐ Client consent given

Save client and start maternal visit

DEVELOPER FOCUS

- Generate a permanent NurtureLink ID offline and link it to a household ID.
- Compare name, date of birth, community and folder/card number before creating a new record.
- Use caregiver consent for a child and client consent for a pregnant adult.

6. Maternal Assessment and Emergency Risks

The pregnancy visit form should not look like the child visit form. It should capture maternal nutrition, ANC progress and danger signs.

Section	Fields
Routine measurements	Height (usually entered once), weight at each visit, weight trend, booking or early-pregnancy BMI, haemoglobin, blood pressure and gestational age.
Nutrition	24-hour dietary recall, maternal dietary diversity, supplements prescribed or given, adherence and barriers to taking supplements.
ANC history	ANC attendance, missed visit, expected delivery date, previous pregnancy complications, gravida and parity.
Danger signs	Severe headache, blurred vision, swelling, vaginal bleeding, severe abdominal pain, convulsions, fever, difficulty breathing and reduced fetal movement.

6.1 Maternal death risks the app should help detect early

The app should be designed around the warning signs of major maternal emergencies, especially severe bleeding, hypertensive disorders, infection and prolonged or obstructed labour. The app should not diagnose these conditions by itself. It should recognize warning information and move the worker quickly to assessment and referral.

- **Haemorrhage:** Heavy bleeding during pregnancy, delivery or after delivery.
- **Hypertensive disorder:** Very high blood pressure, severe headache, blurred vision, swelling or convulsions.
- **Prolonged or obstructed labour:** Labour lasting beyond the approved threshold or signs that the baby is not progressing.
- **Infection or sepsis:** Fever, foul-smelling discharge, severe weakness or other protocol-defined warning signs.

Maternal emergency suspected. Immediate clinical assessment and referral are required. Do not delay referral for routine nutrition counselling.

6.2 Delivery and postpartum fields

- Delivery date and place
- Delivery outcome
- Mode of delivery, when appropriate for the user role
- Heavy bleeding after delivery
- Placenta status, (only when recorded by an appropriately trained worker)
- Blood pressure and convulsions
- Fever or suspected infection
- Severe weakness, fainting or loss of consciousness
- Postpartum follow-up date

The mother must not disappear from the app after delivery. Her status should move from Pregnant to Postpartum/Lactating.

7. Postpartum and Lactating-Mother Assessment

CWC also involves the mother. The child's feeding cannot be understood properly without checking the mother's health, diet and breastfeeding experience.

- Breastfeeding status
- Whether the baby is exclusively breastfed
- Frequency of breastfeeding
- Breastfeeding difficulty or pain
- Mother's 24-hour dietary recall
- Maternal dietary diversity
- Weight or MUAC where required by the approved protocol
- Postpartum danger signs
- Supplements or medicines being taken
- Nutrition counselling provided
- Diet modification agreed with the mother
- Next review date

One woman's record should continue through stages: Pregnant → Delivered → Postpartum/Lactating.

Suggested Interface: Maternal and Postpartum Continuum

The mother's record should continue from pregnancy into delivery, postpartum care and breastfeeding support.

IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Pregnancy visit

11:01 Offline 62%

< **Maternal visit**
Amina Mahama · 24 weeks

MEASUREMENTS

WEIGHT
Enter weight

HAEMOGLOBIN
Enter Hb

BLOOD PRESSURE
_ / _

GESTATIONAL AGE
24 weeks Auto

MUAC
Enter if required by protocol

NUTRITION AND ANC

☐ 24-hour dietary recall

☐ Supplements and adherence

☐ ANC attendance / missed visit

MATERNAL DANGER SIGNS

☐ Severe headache or blurred vision

☐ Swelling of face or hands

☐ Bleeding or severe abdominal pain

☐ Reduced fetal movement

Save maternal visit

Postpartum / lactating visit

11:01 Offline 62%

< **Postpartum / lactating visit**
Amina Mahama · 10 days after delivery

MATERNAL SAFETY

☐ Heavy bleeding

☐ Fever or foul-smelling discharge

☐ Severe headache or blurred vision

BREASTFEEDING

☐ Breastfeeding now

☐ Exclusive breastfeeding

☐ Pain, poor attachment or low milk concern

MOTHER NUTRITION

MATERNAL DIET DIVERSITY
3 of 10 food groups Low

Linked newborn: Baby Rahim
Open newborn

NEXT REVIEW DATE
Select date

Save postpartum visit

DEVELOPER FOCUS

- Show pregnancy-specific measurements, nutrition and danger signs.
- Postpartum care must continue to check bleeding, infection, hypertension and breastfeeding.
- Link the mother directly to the newborn record instead of creating disconnected records.

8. Newborn Assessment

Newborn section	What to capture
Birth details	Date and time of birth, sex, gestational age when known, birth weight and place of delivery.
Risk flags	Low birth weight, preterm birth and reported complications during childbirth.
Breathing at birth	Whether the baby cried or breathed immediately and whether resuscitation was given, when known.
Feeding	Breastfeeding initiation, current feeding, poor feeding and difficulty attaching.
Danger signs	Fast or difficult breathing, fever or low temperature, convulsions, unusual sleepiness, jaundice, cord redness/discharge/bleeding and suspected infection.
Congenital conditions	Congenital anomaly recorded by a qualified professional.
Action	Referral decision, receiving facility and next postnatal visit.

Urgent newborn assessment required: Baby is 3 days old, is feeding poorly and has a low temperature. Immediate referral is recommended.

Low birth weight should be treated as an important risk marker that may be linked to prematurity or poor fetal growth. The app should prioritize the baby for closer follow-up rather than calling low birth weight a single diagnosis.

9. Child Welfare Clinic Growth and Nutrition Assessment

CWC is wider than MUAC and dietary diversity. The child form should combine growth, feeding, supplementation, immunization, counselling and referral.

- Weight
- Length for children measured lying down or height for children measured standing, according to protocol
- MUAC when age-appropriate
- Bilateral oedema
- Weight-for-age
- Length/height-for-age
- Weight-for-length/height
- Growth trend across visits
- Breastfeeding status
- Complementary feeding
- Dietary diversity
- Meal frequency
- Vitamin A status
- Immunization status
- Illness and child danger signs
- Counselling given

- Referral decision
- Next CWC appointment

9.1 Show growth as a trend, not one isolated number

**Weight increased normally since the last visit. / Weight has remained flat across three visits.
/ Weight-for-age is in the severely underweight range.**

The app should use age- and sex-appropriate growth interpretation. A child should not be judged against one general recommended weight.

9.2 Change feeding questions according to age

Age group	Main purpose	Example questions
0–5 months	Exclusive breastfeeding assessment	Only breastmilk? Any water, formula or food? Feeding frequency? Difficulty feeding?
6–23 months	Complementary feeding and continued breastfeeding	Food groups, meal frequency, food texture, quantity, feeding during illness and responsive feeding.
24–59 months	Family diet and growth monitoring	Food variety, meal frequency, illness-related feeding, household food access and growth trend.

The dietary section should lead to an action. It should not stop after showing a score such as 3 out of 8.

**Diet diversity is low. The child did not receive legumes, eggs or vitamin-A-rich foods.
Suggested local options: mashed cowpea, egg or dark-green leaves, where available and appropriate.**

Suggested Interface: Newborn and CWC Pathways

A newborn needs special danger-sign checks, while an older child needs a structured CWC visit.

IN SIMPLE TERMS

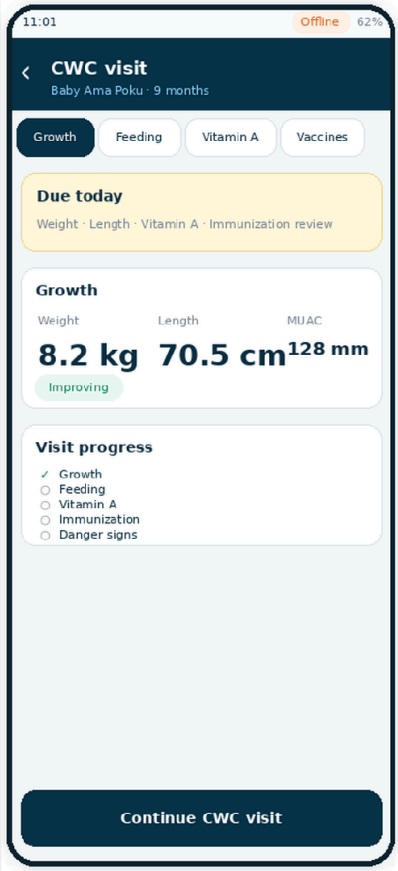
This interface turns the recommendation into a screen the user can understand and act on without guessing.

Newborn assessment



The 'Newborn assessment' screen shows a 'Newborn check' for 'Baby Rahim' at 'Age 3 days'. It features a 'Birth weight 2.3 kg' with a 'LOW BIRTH WEIGHT' warning. Below are sections for 'FEEDING AND WARMTH', 'BREATHING AND ACTIVITY', and 'OTHER NEWBORN CHECKS', each with checkboxes for various symptoms. At the bottom are 'Routine follow-up' and 'Urgent referral' buttons.

CWC visit hub



The 'CWC visit hub' screen shows a 'CWC visit' for 'Baby Ama Poku' at '9 months'. It has tabs for 'Growth', 'Feeding', 'Vitamin A', and 'Vaccines'. The 'Growth' tab is active, showing 'Due today' for 'Weight · Length · Vitamin A · Immunization review'. It displays growth metrics: '8.2 kg', '70.5 cm', and '128 mm' with an 'Improving' status. A 'Visit progress' section shows a checklist with 'Growth' checked and 'Feeding', 'Vitamin A', 'Immunization', and 'Danger signs' unchecked. A 'Continue CWC visit' button is at the bottom.

DEVELOPER FOCUS

- Treat birth asphyxia, congenital anomalies and low birth weight as separate concepts.
- Use clear CWC sections: Growth, Feeding, Vitamin A, Immunization and Danger Signs.
- Show which services are due today and allow the visit to be saved offline.

10. Vitamin A, Immunization and AEFI Tracking

10.1 Vitamin A supplementation

Prototype age rule: under 6 months — not yet eligible; 6–11 months — Blue capsule, 100,000 IU; 12–59 months — Red capsule, 200,000 IU. The next dose date should be calculated from the last recorded dose using a configurable programme interval.

The child record should store:

- Whether Vitamin A was given
- Date given
- Actual dose in IU
- Child's age at the time
- Next dose due
- Reason not given, when applicable

Vitamin A: Child age 9 months | Dose due 100,000 IU | Last dose not recorded | Status: Due today

The interface may show the familiar capsule colour as a visual guide, but it must store the actual dose because packaging can change. The final Ghana schedule must be confirmed before production.

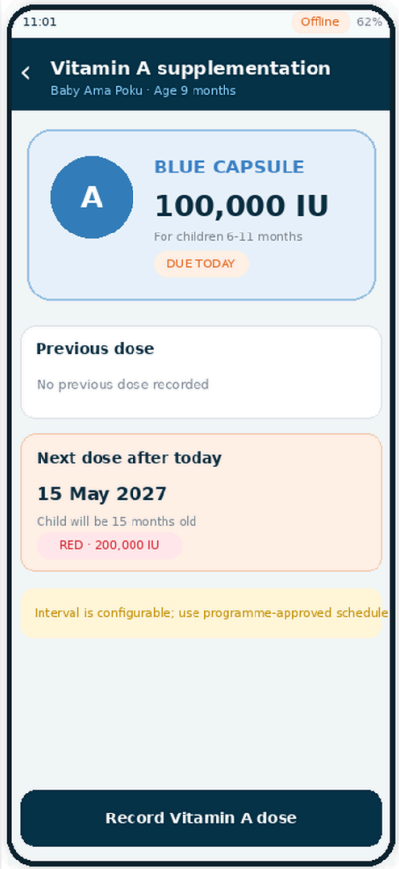
Suggested Interface: Vitamin A Status and Dose Recording

The app should calculate the correct dose from age and remember when the next dose is due.

IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Vitamin A status



Vitamin A supplementation
Baby Ama Poku · Age 9 months

BLUE CAPSULE
100,000 IU
For children 6-11 months
DUE TODAY

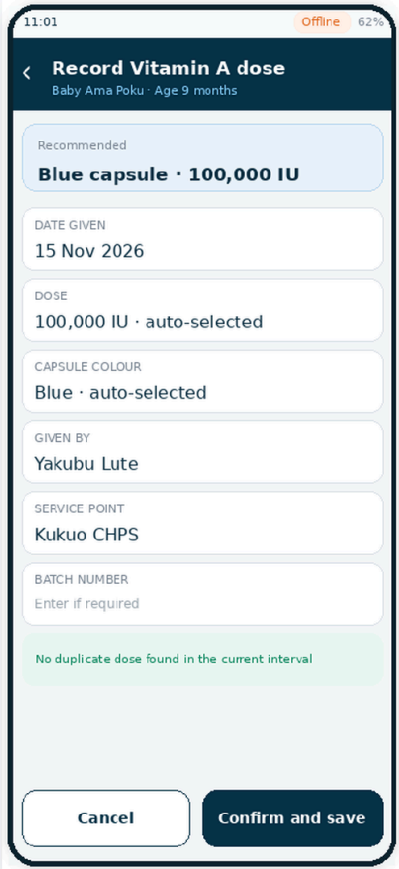
Previous dose
No previous dose recorded

Next dose after today
15 May 2027
Child will be 15 months old
RED - 200,000 IU

Interval is configurable; use programme-approved schedule

Record Vitamin A dose

Record Vitamin A dose



Record Vitamin A dose
Baby Ama Poku · Age 9 months

Recommended
Blue capsule · 100,000 IU

DATE GIVEN
15 Nov 2026

DOSE
100,000 IU · auto-selected

CAPSULE COLOUR
Blue · auto-selected

GIVEN BY
Yakubu Lute

SERVICE POINT
Kukuo CHPS

BATCH NUMBER
Enter if required

No duplicate dose found in the current interval

Cancel **Confirm and save**

DEVELOPER FOCUS

- 6–11 months: Blue capsule, 100,000 IU. 12–59 months: Red capsule, 200,000 IU.
- Calculate the next due date from the recorded dose and recalculate the colour using age at the next visit.
- Warn about duplicate doses, wrong age-dose combinations and invalid dates.

10.2 Immunization tracker

Every child should have an age-based immunization tracker. Pentavalent vaccine is one example, but the system should not be built around only one vaccine.

- Vaccine name
- Dose number
- Recommended age
- Date due
- Date given
- Facility or outreach point
- Health worker who administered it
- Batch or lot number, where required
- Administration site or route, where required
- Status: due, given, missed or overdue
- Reason not given
- Next appointment date

Penta dose due today. / Measles–Rubella dose overdue by 14 days. / Child is up to date for age.

The immunization schedule should be configurable by an authorized administrator. **Do not permanently hard-code the full schedule because Ghana’s Expanded Programme on Immunization can change.**

10.3 Adverse Event Following Immunization (AEFI)

AEFI means a medical event that happens after immunization. It does not automatically mean the vaccine caused the event. The app should record, assess, refer and report according to the approved AEFI protocol.

- Child and client IDs
- Vaccine and dose
- Date and time administered
- Batch or lot number
- Facility or outreach site
- Date and time symptoms started
- Signs and symptoms
- Severity
- Treatment or first aid provided
- Referral and hospitalization status
- Outcome
- Person who reported the event
- Whether the official AEFI form was completed
- Whether the supervisor or district team was notified

A medical event was reported after immunization. This does not prove causation. Follow the approved AEFI assessment, referral and reporting process.

Caregiver counselling after vaccination should explain expected mild reactions, what to do at home, warning signs that require urgent review and the next vaccine appointment.

Suggested Interface: Immunization and AEFI

Immunization must show what is due, what was given and what happened after vaccination.

IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Immunization tracker

11:01 Offline 62%

< Immunization tracker
Baby Ama Poku · Age 9 months

Up to date for age · next visit scheduled

Schedule

Penta 1
Given · 6 weeks GIVEN

Penta 2
Given · 10 weeks GIVEN

Penta 3
Given · 14 weeks GIVEN

Next approved vaccine DUE
Due 15 Dec 2026

Record dose
Vaccine · dose · date · facility · worker
Batch/lot and route/site when required
[Open dose form](#)

Save Immunization review

AEFI report

11:01 Offline 62%

< Report event after immunization
AEFI does not automatically mean causation

Urgent warning signs?
Refer immediately according to approved protocol

VACCINE AND DOSE
Select

DATE/TIME GIVEN
Enter

BATCH/LOT
Enter

SYMPTOMS AND ONSET
Describe event

SEVERITY
Select

HOSPITALIZED
Yes / No

TREATMENT / OUTCOME
Enter

REPORTING STATUS
Supervisor not yet notified Pending

Save, refer and notify

DEVELOPER FOCUS

- Use a configurable Ghana immunization schedule rather than permanently hard-coding every date.
- Track each vaccine dose, date, service point and batch/lot number where required.
- AEFI means an event after immunization; the interface must not claim that the vaccine caused it.

11. Clinical Fields, References and Validation

11.1 Do not show blank values as zero

Do not display 0.0 kg, 0.0 g/dL or 000 mm before the worker enters a value. A zero can look like a real measurement.

Field	Recommended placeholder
Weight	Enter weight
Haemoglobin	Enter haemoglobin
MUAC	Enter MUAC
Blood pressure	Enter systolic and diastolic values

11.2 Validate unusual or impossible values

MUAC 850 mm appears outside the expected range. Please recheck the measurement.

Validation should prevent typing mistakes without blocking a genuine unusual case. The worker should be able to confirm an unusual value and add a note.

11.3 Haemoglobin interpretation

After entering haemoglobin, the app should show whether the value is below the approved reference for the pregnancy stage and whether it is falling across visits.

Haemoglobin: 9.6 g/dL | Below the approved reference for this pregnancy stage | Falling across three visits | Follow the approved ANC protocol

The wording must be consistent. The same value should not be called mild anaemia on one screen and approaching anaemia on another. Clinical thresholds and actions must be validated by Ghana Health Service or approved Ghanaian clinicians.

11.4 Weight interpretation

- **Pregnancy:** Show current weight, change since last visit, change since booking or first ANC visit, gestational age and trend. Do not show one general recommended pregnancy weight.
- **Child:** Use age, sex, length/height, weight, MUAC and growth history. Do not show one general recommended child weight.

12. Explainable AI and Daily Prioritization

The “Why NurtureLink ranked this client” section is one of the strongest parts of the prototype. Keep it and make it available for every priority decision.

- What changed

- How the value changed
- What approved threshold or rule was crossed
- Why the client moved up the list
- What action is recommended
- How urgent the action is
- When follow-up is due

MUAC declined from 117 mm to 108 mm. The child has not gained weight across two visits. Diet diversity is 1 out of 8. This is why the child is first on today's list.

12.1 Priority reasons the app should understand

- Current danger sign
- Worsening trend across visits
- Missed CWC, ANC or postnatal visit
- Referral not confirmed
- Falling haemoglobin
- Poor or flat child growth
- Low MUAC or oedema
- Low dietary diversity
- Vitamin A due
- Immunization due or overdue
- Newborn risk
- Old unsynchronized record that blocks reporting

13. Local-Food Plans, Voice Counselling and Delivery

The local-food feeding plan and voice counselling are practical strengths. Keep them, but make the assumptions and delivery history clearer.

- Use foods available in the community and season.
- Consider household affordability.
- Respect allergies, preferences and cultural or religious restrictions.
- Change advice according to pregnancy, lactation or child age.
- Show why each food was suggested.
- Allow the worker to edit, remove or replace recommendations.
- Separate nutrition support from clinical treatment and prescribed supplements.

NurtureLink suggests. The health worker reviews, edits and approves. The caregiver receives the final message.

If the app displays nutrient percentages, it should explain the assumed portion sizes, number of meals, life stage and food-composition source. Do not present very precise percentages without showing the assumptions.

13.1 Record how counselling was delivered

- Played aloud during the visit
- In-person verbal counselling
- Bluetooth to a feature phone
- WhatsApp, queued until online
- Worker-recorded local-language voice note

The app should confirm consent before sending health information through WhatsApp or Bluetooth and should send only the minimum necessary information.

Suggested Interface: Explainable AI and Counselling Delivery

The worker should see why a client is prioritized, review the advice and record how it was delivered.

IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Why this client is prioritized

11:01 Offline 62%

< Why this client is first

Baby Labf · 9 months

EVIDENCE

MUAC fell from 117 mm to 108 mm
No weight gain across 2 visits
Diet diversity is 1 of 8

REFERRAL STATUS

Urgent referral is still awaiting confirmation
Follow-up is due today

WHAT THE APP RECOMMENDS

Call the caregiver or facility
Confirm whether the child was seen
Record the outcome and next follow-up

HUMAN DECISION

The worker can confirm the recommendation
or record another clinical judgement.

Open referral follow-up

Home

Referrals

Sync

Profile

Counselling delivery

11:01 Offline 62%

< Deliver counselling

Approved plan for Amina

LANGUAGE

English Dagbani Other local language

DELIVERY METHOD

Play aloud now
Bluetooth to feature phone
WhatsApp when online
Record worker voice note
In-person verbal counselling

CONSENT AND RECORD

☐ Caregiver/client agrees to selected method
Method, date, time and worker will be saved

PRIVACY

Only the minimum necessary health information
should be sent through digital channels.

Mark as delivered

Home

Referrals

Sync

Profile

DEVELOPER FOCUS

- Explain the changing measurement, missed service or threshold that caused the alert.
- Keep the health worker in control: review, edit, approve or reject the suggestion.
- Record whether counselling was played aloud, sent by Bluetooth/WhatsApp or delivered in person.

14. Referral and Follow-Up History

The referral process is already strong, but “Confirm seen” should collect enough information to create a proper history.

- Confirmation date and time
- Facility
- How confirmation was obtained
- Who confirmed it — initials or staff ID
- Outcome
- Treatment or follow-up instruction, when available
- Next follow-up date
- Worker or supervisor who recorded the confirmation

An active referral already exists for this client. Update the current referral or create a new referral episode?

Possible confirmation methods:

- Facility feedback
- Caregiver phone call
- Home follow-up
- Referral slip
- Supervisor or district confirmation

Serious maternal or newborn danger signs should bypass routine counselling and move directly to urgent referral, facility contact and confirmation tracking.

Suggested Interface: Referral Confirmation and Outcome

"Confirm seen" should create a complete referral history, not just place a tick beside the client.

IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Confirm facility attendance

The interface is a mobile app screen titled "Confirm facility attendance" for "Baby Latif - Tamale West Hospital". It features a status bar at the top with the time 11:01, "Online" status, and 62% battery. The main content area includes a yellow header "Referral awaiting confirmation" with the text "Issued 12 Nov 2026 · urgent". Below this are several input fields: "CONFIRMATION DATE" (Select), "CONFIRMATION TIME" (Enter), "HOW CONFIRMATION WAS OBTAINED" (Facility feedback / caregiver call / home visit), "WHO CONFIRMED" (Initials or staff ID), "FACILITY OUTCOME" (Seen / admitted / treated / referred onward), "OUTCOME NOTE" (Optional details), and "NEXT FOLLOW-UP DATE" (Select). A large green button at the bottom is labeled "Save referral outcome".

DEVELOPER FOCUS

- Record date, facility, confirmation method and confirmer initials or staff ID.
- Capture the outcome and the next follow-up date.
- Prevent duplicate active referrals for the same client and clinical episode.

15. Offline Saving, Synchronization and DHIMS2 Reporting

The app must separate three actions clearly:

1. Saving a record safely on the device
2. Synchronizing the record to the server
3. Completing and submitting the monthly DHIMS2 report

A worker can have every record synchronized and still not have completed the monthly report.

15.1 Offline and online status

State	Recommended message
Offline	Working offline. Five records are encrypted and safely stored on this device.
Online and complete	All records synchronized successfully. Last successful sync: date and time.
Online with problem	Four records synchronized. One record needs correction.

15.2 Monthly reporting centre

The monthly DHIMS2 reporting function should not be hidden inside Profile. It should be visible from the worker and supervisor dashboards.

November monthly report | 87% complete | 6 records not synchronized | 2 records need correction | Deadline: 15 December

The current project requirement is a deadline on the 15th of each month. The developer should make this date configurable and confirm the final facility/district rule with Ghana Health Service before deployment.

Reporting stages:

Draft → Complete → Reviewed → Submitted → Accepted

15.3 Required reminders

- Reminder before the deadline
- Stronger reminder a few days before the deadline
- Reminder on the deadline
- Overdue alert after the deadline
- Alert when missing data or unsynchronized records prevent submission
- Alert when the report is submitted but not yet accepted

Monthly report overdue. Three records are unsynchronized and one required field is missing.

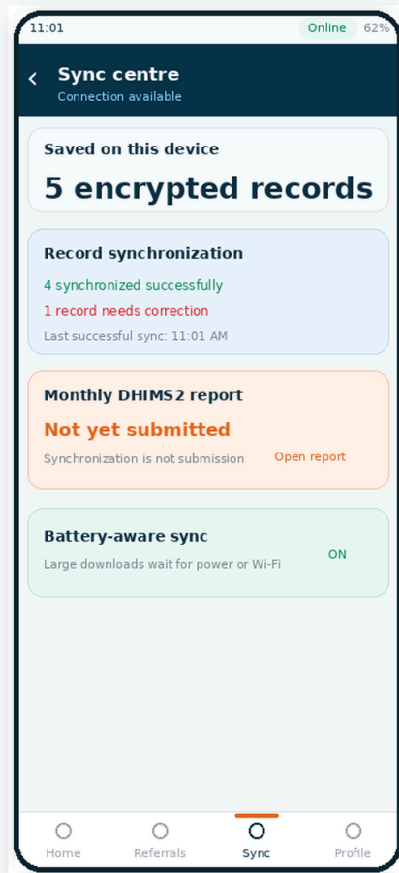
Suggested Interface: Sync and Monthly DHIMS2 Reporting

Saving locally, synchronizing records and submitting the monthly report are three different actions.

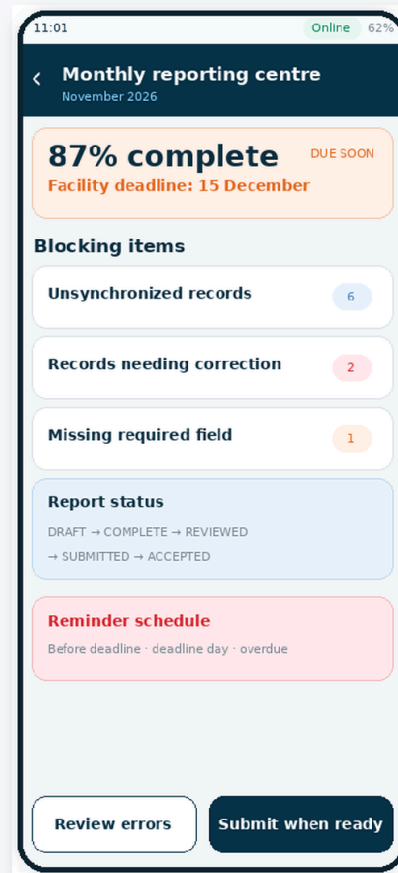
IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Sync centre



Monthly reporting centre



DEVELOPER FOCUS

- Show exactly which records are saved, synchronized or blocked by an error.
- Display report completeness, deadline, missing fields and submission stage.
- Send reminders before the deadline, on the deadline and after the report becomes overdue.

16. Data-Quality Alerts

The app should not wait for the supervisor to manually discover every problem. It should create simple, actionable data-quality alerts.

- Missing NurtureLink ID, household ID or folder number
- Missing consent
- Impossible or unusual clinical value
- Duplicate client
- Duplicate active referral
- Old unsynchronized record
- Referral overdue
- CWC, ANC or postnatal visit overdue
- Pregnancy record without gestational age or expected delivery date
- Child record without date of birth or estimated age
- Vaccine recorded without dose/date/batch details where required
- AEFI record incomplete
- Monthly total does not match individual client records
- Required report field missing

Data-quality alert: Two clients may be duplicates. One referral is overdue. Three records are blocking the monthly report.

17. Supervisor Dashboard and Permissions

The supervisor dashboard should not be only a larger health-worker dashboard. It should answer five simple questions:

- Which community has the highest unresolved risk?
- Which worker needs support?
- Which referrals are still open?
- Which monthly reports are incomplete?
- Where are maternal, newborn, nutrition or immunization problems increasing?

Supervisor section	What it should show
District summary	Active clients, pregnant women, lactating mothers, newborns, children under five, urgent cases, open referrals and referral completion.
Zone comparison	Follow-up load, urgent cases, overdue CWC/ANC visits, immunization gaps, unsynced records and reporting status.
Field officers	Assigned zone, caseload, urgent cases, referrals awaiting confirmation, last sync time and monthly report status.
Nutrition and CWC	Growth faltering, severe underweight/wasting alerts, dietary diversity, Vitamin A coverage and children overdue for CWC.
Maternal/newborn	Falling haemoglobin, maternal danger alerts, low-birth-weight newborns and newborn danger alerts.
Data quality	Duplicates, impossible values, missing consent, missing IDs, incomplete reports and old unsynced records.

17.1 Supervisor permissions

The supervisor should mainly:

- Review performance and workload
- Monitor referrals and follow-up
- Review monthly reports
- Identify data-quality problems
- Send feedback
- Escalate unresolved risks
- View worker caseloads in read-only mode

The supervisor should not silently:

- Create a visit under a worker's name
- Edit clinical data without an audit trail
- Confirm a referral without recording the source
- Change a monthly report after submission without recording the correction

Any supervisor correction should record who changed the data, what changed, when it changed and why.

18. Mortality and Outcome Indicators

The mortality idea is useful, especially for the supervisor dashboard. However, the app must clearly separate official population or facility indicators from NurtureLink's own client counts.

Indicator	Display rule
Maternal mortality ratio	Show per 100,000 live births, with location, period, source and last updated date.
Neonatal mortality rate	Show per 1,000 live births, with location, period, source and last updated date.
Under-five mortality rate	Show per 1,000 live births, with location, period, source and last updated date.
Severe malnutrition	Show cases, rates or coverage using a clearly defined denominator.
Maternal anaemia	Show number or proportion, pregnancy stage where relevant, location and reporting period.
Referral completion	Show referrals completed within the required time, with numerator and denominator.

Deaths recorded among NurtureLink clients: 2

18.1 Do not use mortality alone to score one health worker

Mortality is affected by many things outside one worker's control, including referral transport, facility capacity, poverty and late care-seeking. It can provide context, but it should not be used alone to say that one worker is performing well or poorly.

Better worker-level performance indicators include:

- Timely completion of urgent follow-up
- Referral confirmation rate
- Children brought back for overdue CWC or immunization
- Vitamin A coverage
- ANC and postnatal follow-up completion
- Data completeness and accuracy
- Timely synchronization
- Monthly report submitted on time
- Counselling delivery and documented follow-up

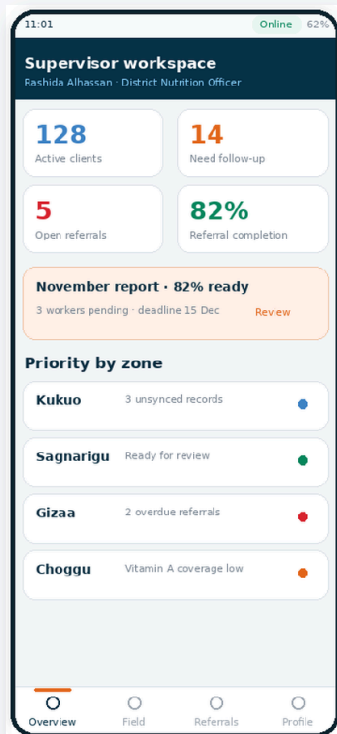
Suggested Interface: Supervisor Oversight and Outcomes

The supervisor needs a district view that shows where support, correction or escalation is required.

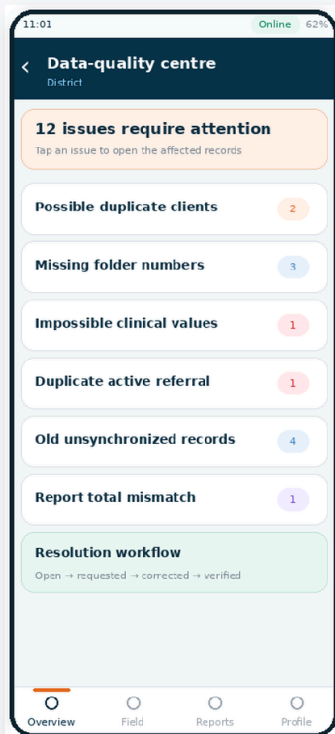
IN SIMPLE TERMS

This interface turns the recommendation into a screen the user can understand and act on without guessing.

Supervisor overview



Data-quality centre



Outcomes



DEVELOPER FOCUS

- Show workload, open referrals, report readiness and priority by zone.
- Make data-quality alerts actionable and traceable from open to corrected to verified.
- Separate official mortality indicators from NurtureLink operational counts and do not judge a worker by mortality alone.

19. Privacy, Security and Audit Trail

The shared-device PIN is a good beginning. Because the device holds maternal and child health information, the following protections should also be built in.

- Automatic screen lock
- PIN retry limit
- Encrypted offline records
- Role-based access
- Consent history
- Audit history for view, edit, referral, counselling and report actions
- Minimum necessary personal information
- Hidden sensitive notification previews
- Ability to deactivate a lost or reassigned device when connectivity becomes available

Separate consent may be needed for:

- Storing and synchronizing health information
- Sending WhatsApp counselling
- Sending Bluetooth audio
- Recording and storing voice counselling

20. Complete User Journeys

20.1 Health-worker journey

1. Log in to the Health-worker workspace.
2. See offline/online status and the active tab.
3. Review who needs action today and why.
4. Search for an existing household or client before registering a new one.
5. Register the correct client type and capture consent.
6. Load the correct maternal, lactating, newborn or child form.
7. Record measurements, diet, supplementation, immunization and danger signs.
8. See a simple explanation of risk and trend.
9. Approve counselling, monitoring or referral.
10. Deliver local-language guidance and record the method.
11. Track referral confirmation and follow-up.
12. Save offline and synchronize when connectivity returns.
13. Resolve errors and complete the monthly DHIMS2 report.

20.2 Supervisor journey

1. Log in to the Supervisor workspace.
2. Review district, zone and worker summaries.
3. See unresolved maternal, newborn, child, nutrition, immunization and referral alerts.

4. Review data-quality and unsynchronized-record problems.
 5. Open a worker's caseload in clearly labelled read-only mode.
 6. Send feedback or escalate unresolved risks.
 7. Review zone reports and validation errors.
 8. Approve or submit the monthly report according to assigned permissions.
 9. Monitor official outcome trends and clearly labelled NurtureLink operational counts.
- and under-five deaths in Northern Ghana.